

DENTING APP

Stunting Detection Application to Help Parents
in Detecting the Risk of Stunting in Children.

CH2-PS392



DENTING

D E T E K S I S T U N T I N G

OUR TEAM



Zuama (ML)



Bayu (ML)



Wawan (MD)



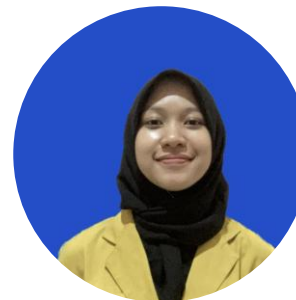
Faiz (CC)



Dika (CC)



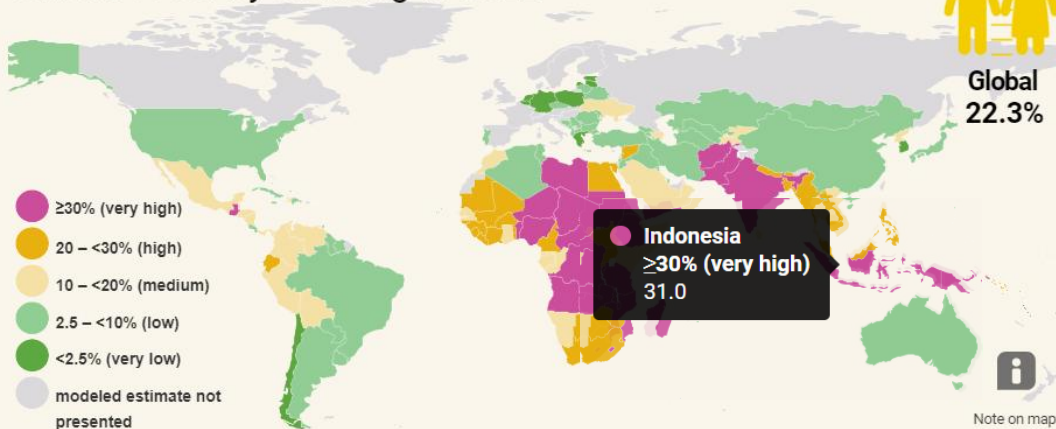
Wahyu (MD)



Balqist (ML)

High Stunting Prevalence in Indonesia

In 28 countries, at least 30 per cent of children were still affected by stunting in 2022



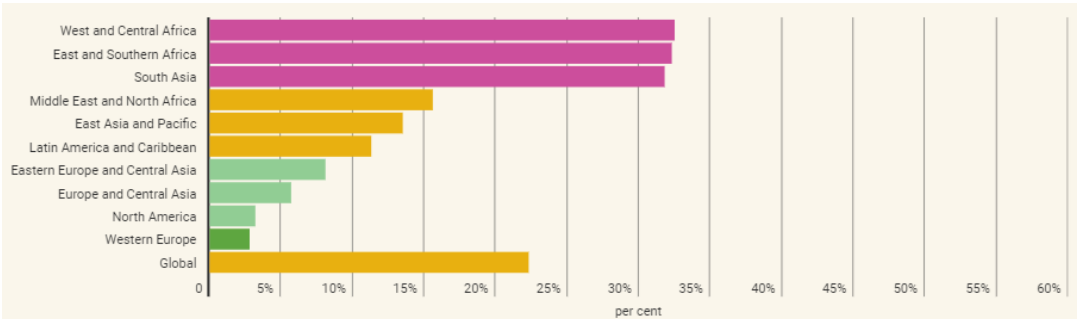
Percentage of children under 5 affected by stunting, by country, 2022

Note: *Asterisks denote the most recent country data point (e.g., from household surveys) used to generate the modelled stunting and overweight estimates is from before the year 2000; interpret with caution.



Source: BUKU SAKU Hasil Survei Status Gizi Indonesia (SSGI) 2022

High Stunting Prevalence in Indonesia



Percentage of children under 5 affected by stunting (%), by UNICEF region, 2022

Source: UNICEF, WHO, World Bank Group Joint Child Malnutrition Estimates, 2023 Edition



Source: BUKU SAKU Hasil Survei Status Gizi Indonesia (SSGI) 2022

Government Measures and Targets



DINAS KOMINFO
PROVINSI JAWA TIMUR

HOME | PROFIL | PROGRAM | LAPORAN | LAYANAN | REGULASI | PID | PUBLIKASI

Selasa, 19 Desember 2023

[Beranda](#) / [Berita](#)

Program 1-10-100, Langkah Strategis Lamongan Turunkan Stunting

Ditunggah pada : 29 Agustus 2023 8:36:05 215

Sumber Foto: Disdikmabli Kabupaten Lamongan

Jatim Newroom - Pemerintah Kabupaten Lamongan segera mengimplementasikan program 1-10-100 yang menjadi langkah strategis terkini percepatan penurunan stunting di Kabupaten Lamongan mulai September mendatang.

Berita Terkait

- Pemkab Lamongan Serahkan Satu Unit Mobil...**
18 Desember 2023
- Pemkab Lamongan Pastikan JAMULA Ruas Sak...**
18 Desember 2023
- GELOBA CINTA dan PUSYANGATRA Terus Digen...**
15 Desember 2023
- Inilah Tantangan Pelaksanaan Program Ker...**
13 Desember 2023
- Asisten I Setdaprov Jatim Paparkan Pelak...**
13 Desember 2023

Impacts of High High Stunting Prevalence

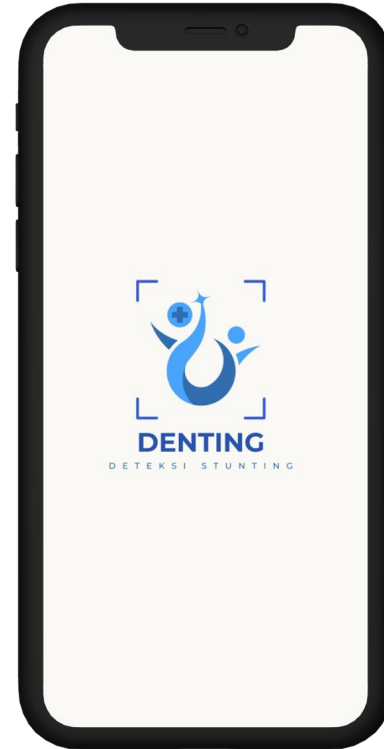
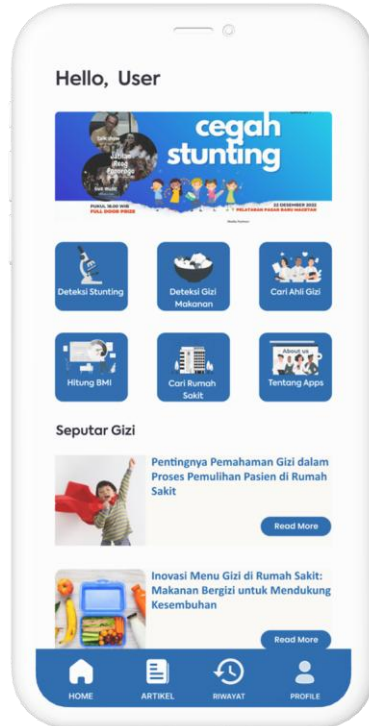
Cognitive and Developmental Challenges

Healthcare Burden

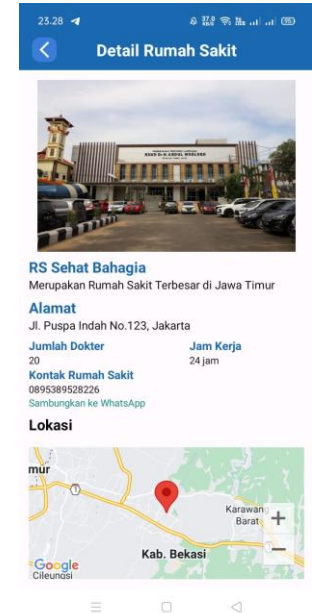
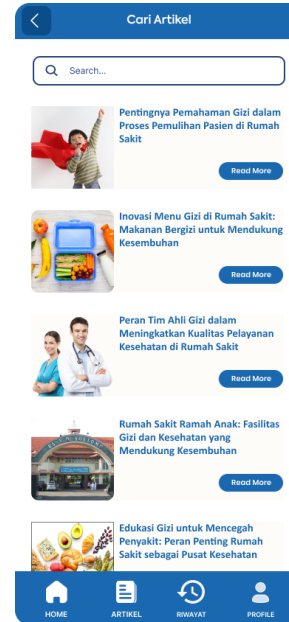
Economic Consequences

Inter-generational Cycle

Product Introduction



Features



Advantages of Denting



- Has many features
- Between one feature and another feature are correlated
- User friendly UI
- Can be an easy measurement for mothers and

App Comparation of Denting

Comparison Point	Denting	Gizi Nusantara
App runs well	✓	✓
Intuitive UI/UX	✓	✓
GPS Feature	✓	x
Stunting Detection	✓	x
Food Nutrition Detection	✓	x
Nutritionists	✓	✓



Machine Learning Path

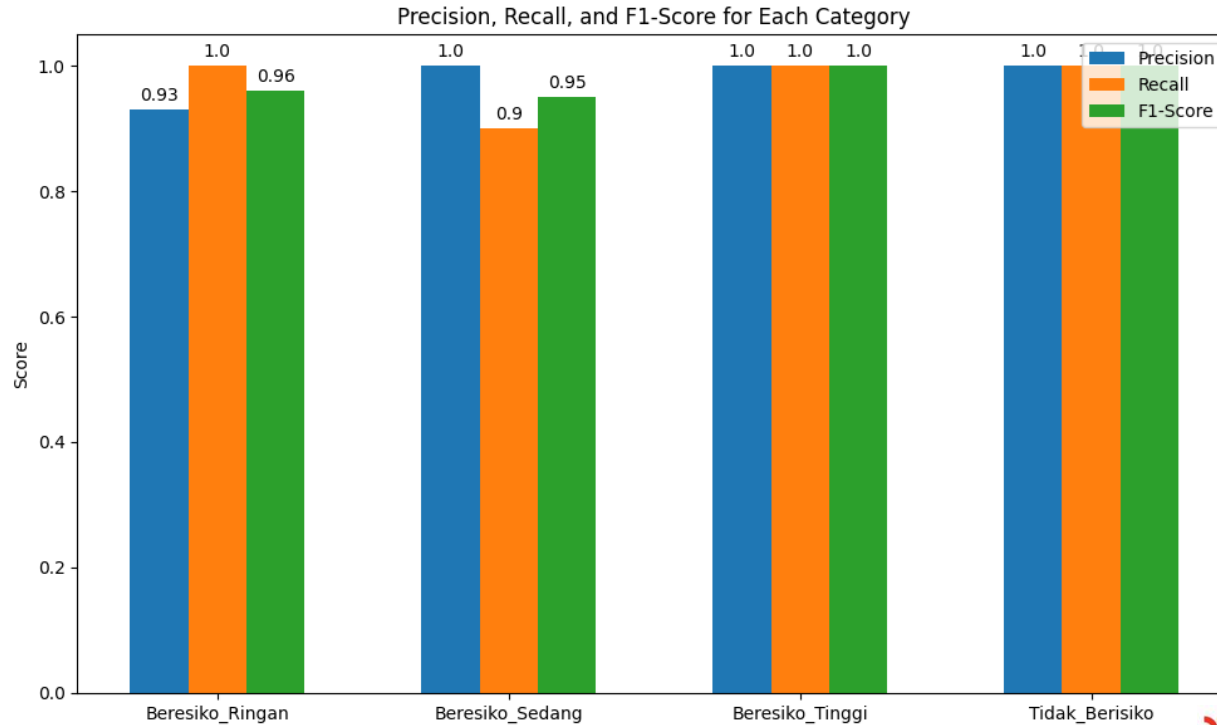
Tools



Stunting Detection Datasets

Berat_Badan	Jenis_Kelamin	Tinggi_Badan	Usia(bulan)	Risiko_Stunting
12	Laki-laki	80	24	Tidak_Berisiko
10	Perempuan	75	18	Beresiko_Ringan
15	Laki-laki	85	30	Tidak_Berisiko
9	Perempuan	70	15	Beresiko_Tinggi
11	Laki-laki	78	22	Beresiko_Ringan
8	Perempuan	68	14	Beresiko_Tinggi
18	Laki-laki	90	36	Tidak_Berisiko
8	Perempuan	65	12	Beresiko_Tinggi
16	Laki-laki	88	32	Tidak_Berisiko
10	Perempuan	72	20	Beresiko_Sedang

Stunting Detection Model Training using Decision tree



Detection Risk Stunting

Masukkan Jenis Kelamin (Laki-laki/Perempuan): Laki-laki

Masukkan Tinggi Badan (cm): 50

Masukkan Berat Badan (kg): 10

Masukkan Usia (bulan): 30

Input values:

Jenis_Kelamin	Tinggi_Badan	Berat_Badan	Usia
0	0	50.0	10.0
			30

Predicted Risk: ['Beresiko_Ringan']

Nutrition Detection

Datasets Training and Validation

```
# Corrected import statement for ImageDataGenerator
from tensorflow.keras.preprocessing.image import ImageDataGenerator  Import "tensorflow.keras.preprocessing.image" could not be resolved

# Example of data augmentation with the corrected import
training_datagen = ImageDataGenerator(
    rescale=1. / 255,
    rotation_range=40,
    horizontal_flip=True,
    shear_range=0.2,
    zoom_range=0.2,
    fill_mode='nearest'
)

validation_datagen = ImageDataGenerator(rescale=1.0 / 255.)
```

✓ 0.0s

Python

```
train_generator=train_datagen.flow_from_directory(directory=train_dir,
                                                    target_size=(128,128),
                                                    batch_size=32,
                                                    shuffle=True,
                                                    class_mode='categorical')

validation_generator=validation_datagen.flow_from_directory(directory=validation_dir,
                                                            target_size=(128,128),
                                                            batch_size=32,
                                                            shuffle=True,
                                                            class_mode='categorical')
```

✓ 0.0s

Python

Found 7760 images belonging to 5 classes.
Found 1940 images belonging to 5 classes.

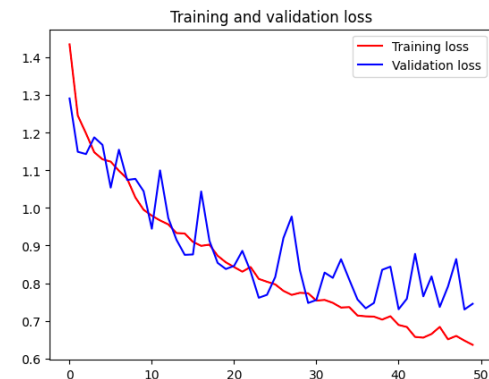
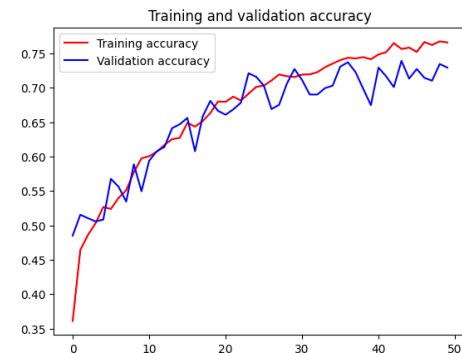
Nutrition Detection Model Training

```

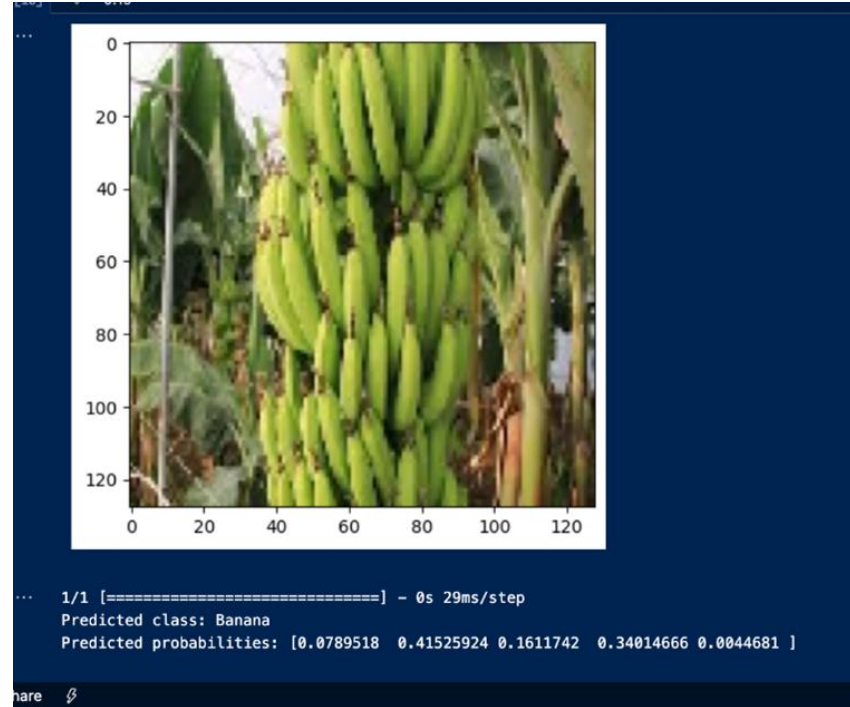
Epoch 1/50
243/243 [=====] - 42s 169ms/step - loss: 1.4351 - accuracy: 0.3610 - val_loss: 1.2913 - val_accuracy: 0.4851
Epoch 2/50
243/243 [=====] - 41s 169ms/step - loss: 1.2458 - accuracy: 0.4643 - val_loss: 1.1494 - val_accuracy: 0.5155
Epoch 3/50
243/243 [=====] - 41s 169ms/step - loss: 1.1983 - accuracy: 0.4860 - val_loss: 1.1434 - val_accuracy: 0.5103
Epoch 4/50
243/243 [=====] - 42s 173ms/step - loss: 1.1483 - accuracy: 0.5032 - val_loss: 1.1878 - val_accuracy: 0.5057
Epoch 5/50
243/243 [=====] - 43s 176ms/step - loss: 1.1295 - accuracy: 0.5267 - val_loss: 1.1678 - val_accuracy: 0.5082
Epoch 6/50
243/243 [=====] - 41s 170ms/step - loss: 1.1233 - accuracy: 0.5238 - val_loss: 1.0541 - val_accuracy: 0.5675
Epoch 7/50
243/243 [=====] - 41s 167ms/step - loss: 1.0989 - accuracy: 0.5399 - val_loss: 1.1550 - val_accuracy: 0.5562
Epoch 8/50
243/243 [=====] - 40s 163ms/step - loss: 1.0779 - accuracy: 0.5510 - val_loss: 1.0745 - val_accuracy: 0.5345
Epoch 9/50
243/243 [=====] - 40s 163ms/step - loss: 1.0279 - accuracy: 0.5772 - val_loss: 1.0774 - val_accuracy: 0.5887
Epoch 10/50
243/243 [=====] - 40s 164ms/step - loss: 0.9953 - accuracy: 0.5974 - val_loss: 1.0446 - val_accuracy: 0.5495
Epoch 11/50
243/243 [=====] - 40s 163ms/step - loss: 0.9795 - accuracy: 0.6005 - val_loss: 0.9447 - val_accuracy: 0.5938
Epoch 12/50
243/243 [=====] - 41s 169ms/step - loss: 0.9669 - accuracy: 0.6070 - val_loss: 1.1000 - val_accuracy: 0.6077
Epoch 13/50
...
Epoch 49/50
243/243 [=====] - 43s 177ms/step - loss: 0.6472 - accuracy: 0.7675 - val_loss: 0.7301 - val_accuracy: 0.7345
Epoch 50/50
243/243 [=====] - 42s 174ms/step - loss: 0.6361 - accuracy: 0.7660 - val_loss: 0.7452 - val_accuracy: 0.7294

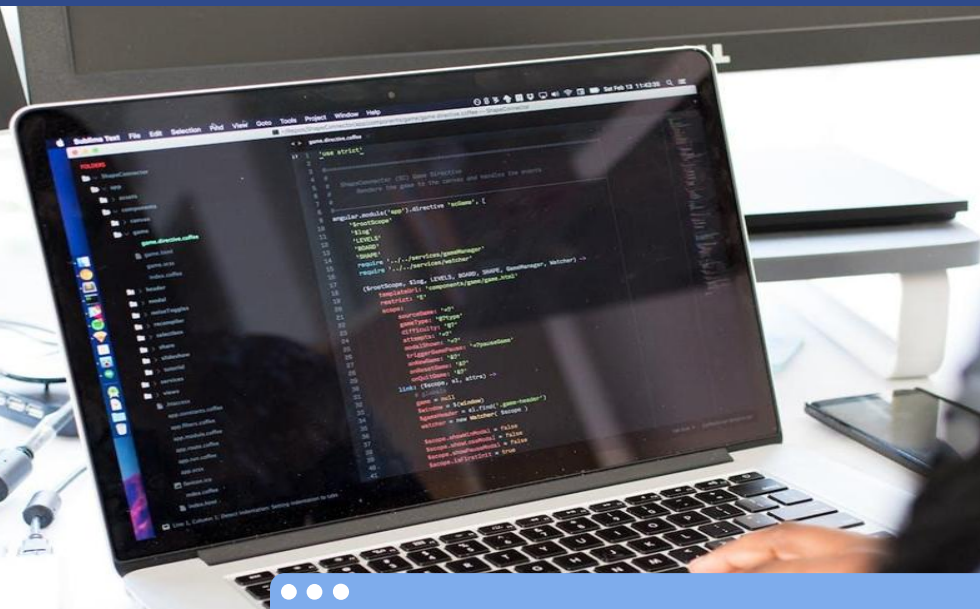
```

Output is truncated. View as a [scrollable element](#) or open in a [text editor](#). Adjust cell output [settings](#)...



Predict Image





Mobile Development Path

WHAT WE DO



Design UI/UX

Building the UI/UX design using Figma



Integrating API

Integrating the Login /Register, Food Detect, and Stunting Detection API to the app



Building the app

Building the App using Kotlin Language in Android Studio

Tools



Android Studio



Kotlin



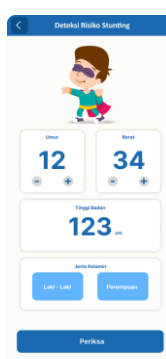
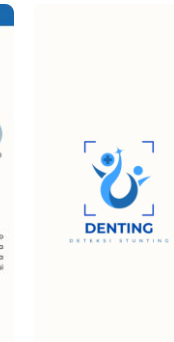
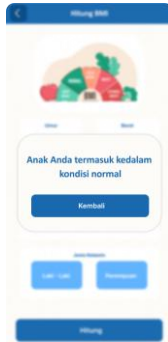
Figma



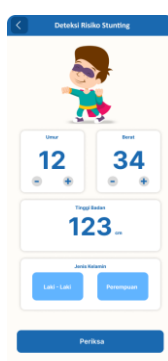
Firebase



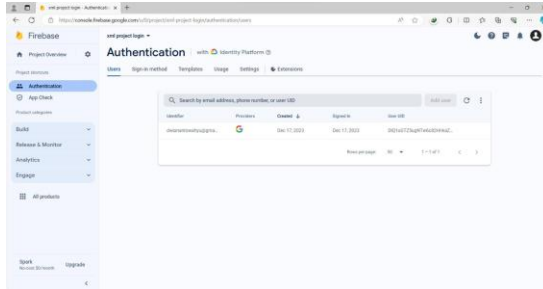
Maps



In App Frontend

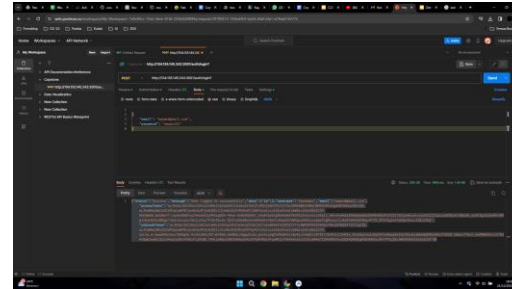


User Authentication



Firestore

Using the firebase to
login/register via google
account



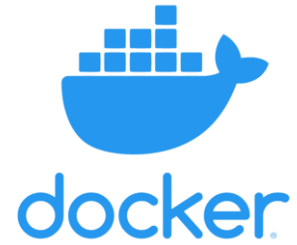
API

Using the API to
login/register the account
using email/password

Cloud Computing Path



Tools



Model deteksi stunting

<https://dentingmodell-asuptnnvoa-as.a.run.app>

```
from flask import Flask, request, jsonify
from flask_cors import CORS
from sklearn.preprocessing import StandardScaler
from sklearn.tree import DecisionTreeClassifier
import joblib

app = Flask(__name__)
CORS(app)

# Load the pre-trained model
model = joblib.load('decision_tree_model.pkl')

@app.route("/")
def welcome():
    return "Welcome to the model prediction API!"

@app.route('/predict', methods=['POST', 'OPTIONS'])
def predict():
    # Get data from the request
    data = request.get_json()

    # Ensure that the expected features are present in the input data
    expected_features = ["Berat_Bahan", "Jenis_Kelamin", "Tinggi_Bahan", "Usia"]
    if not all(feature in data for feature in expected_features):
        raise ValueError("Missing required features in input data")

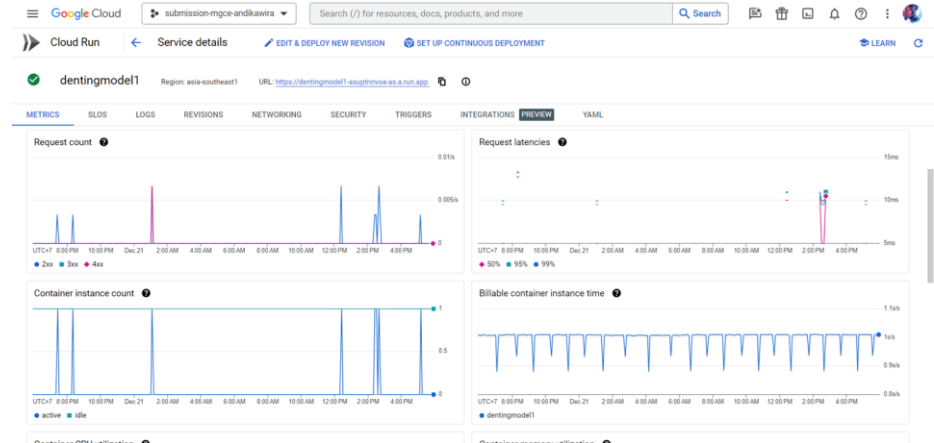
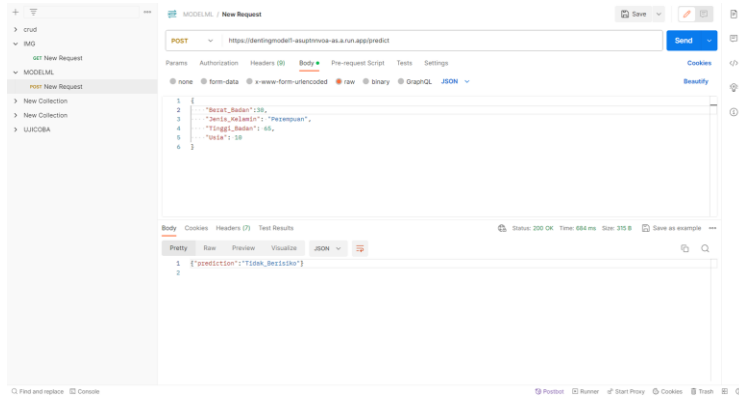
    # Convert "jenis_kelamin" to numeric (0 for 'laki-laki', 1 for 'perempuan')
    data['jenis_kelamin'] = 0 if data['jenis_kelamin'] == 'laki-laki' else 1

    # Create a list from the input data
    input_data = list(data.values())

    # Make prediction
    prediction = model.predict(input_data)

    # Prepare response
    response = {'prediction': prediction}

    # Convert prediction to list if necessary
```



Model image classification

<https://modelimageclass-asuptnnvoa-as.a.run.app>

```
from flask import Flask, request, jsonify
from flask_cors import CORS
import tensorflow.keras
import tensorflow.keras.models
import tensorflow.keras.preprocessing.image
import numpy as np

app = Flask(__name__)
CORS(app)

def load_model_func():
    model_path = 'hail_model_fix_jadi.h5'
    model = tensorflow.keras.models.load_model(model_path)
    return model

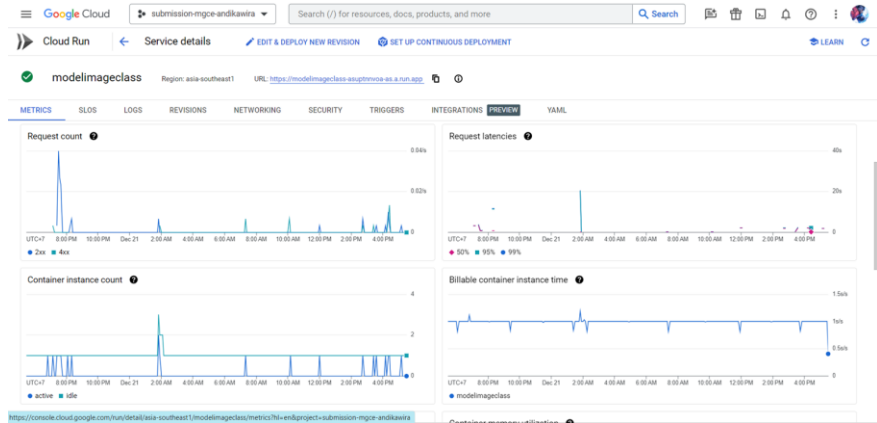
def predict(request):
    # Getting the image from the POST request
    img_data = request.files['image'].read()
    img = image.img_to_array(img_data).astype(np.float32)
    img_array = np.expand_dims(img, axis=0)
    img_array /= 255.0

    # Load the model
    model = load_model_func()

    # Make prediction
    predictions = model.predict(img_array)
    predicted_class = np.argmax(predictions[0])

    # Getting class labels from the generator
    class_labels = ['Apple', 'Banana', 'Grape', 'Mango', 'Strawberry']

    # Predict the class label
```



POST <https://modelimageclass-asuptnnvoa-as.a.run.app/api/predict> Send

Params Authorization Headers Body Pre-request Script Tests Settings Cookies

Key	Value	Description
image	File image.jpg	
Key	Text Value	Description

Body Cookies Headers (7) Test Results Status: 200 OK Time: 19.89 s Size: 311 B Save as example

```
1 {
2   "predicted_class": "Grape"
3 }
```

BUSINESS PLAN (SWOT)



- S : Have many features
- W : <90 target acc (80-89 acc)
- O : High stunting rate Indonesia (>21%)
- T : Introduction to society

BUSINESS PLAN (PARTNERSHIP)



Hospital



Government



BRAND NAME
Tagline Here

Food Brand



Nutritionist

FUNDING ALLOCATION (10K \$ or IDR 155M)

Category	Proportions	Budget (max) in USD
App Development and Team Salary	40%	\$4.000 / Rp 62.000.000
Market Research, Marketing Campaign, and Sales	20%	\$2.000 / Rp 31.000.000
Localization and Translation	10%	\$1.000 / Rp 15.500.000
Future App Development / R&D, Other Expenses	20%	\$2.000 / Rp 31.000.000
Partnership Expenses	10%	\$1.000 / Rp 15.500.000



Thanks!!!

